

## Slides 6.14: Build the Master Flowchart

Lesson 6.14, one day, 13 slides. Slide 5 (the example chart) stays up while teams work.

### Slide 1: Do now

- Think about the pasta lab or the quesadilla lab.
- Write the one moment your station was standing around waiting.
- What were you waiting for? Notes: Take four answers, one per station. Nearly always the same three: the water, the oven, the teacher. Circle them.

### Slide 2: Those are not accidents

- Those are the steps everything else waits on.
- A real kitchen builds the clock around them.
- Today you build the clock. Notes: Then the frame: "I ran four kitchens in one period off a flowchart on one page, in five-minute blocks, written so a substitute could run it. That is what you are making."

### Slide 3: Eight blocks, five minutes each, forty minutes

- 0 to 5 SETUP
- 5 to 10 PREP
- 10 to 15 COOK START
- 15 to 20 COOK
- 20 to 25 KEY CHECK
- 25 to 30 FINISH AND PLATE
- 30 to 35 SERVE AND EAT
- 35 to 40 CLEAN UP Notes: Copy all eight labels onto the builder. Five phases: setup, prep, cook, serve and eat, clean.

### Slide 4: The rule that makes a flowchart work

- **Every block has something for every person.**
- Nobody has a blank block.
- If your station finishes early, that block says SUPPORT DUTY and names who you help.

### Slide 5: What a finished chart looks like

- 0 to 5: **CRITICAL PATH, pot of hot water covered on HIGH.** Sanitation. Bin checked. (Head Cook, Prep Cook, all)
- 5 to 10: Watch for a hard boil. Boxes and jars opened. 30 plates counted. (Head Cook, Prep Cook, Plate Builder)
- 10 to 15: Water boils. Pasta in. Stir. **8-minute timer.** (Head Cook, Second Cook, Time Keeper)
- 15 to 20: Stir every 2 minutes. Sauce lined up. **Helper Cook on SUPPORT DUTY.** (Second Cook, Toppings Chief, Helper Cook)

- 20 to 25: **KEY CHECK.** Teacher drains at minute 20. Sauce in. Parmesan on. (Station Manager, Head Cook)
- 25 to 30: Pot to the table on a trivet. **Plates down at 29.** Allergen box read aloud. (all)
- 30 to 35: Class served first, then teammates, then us. (all)
- 35 to 40: Four clean zones. Station check signed. (all) Image: the same eight blocks drawn as one tall column with a red circle on block one and a star on block five. Notes: Leave this up. Point at block 20 to 25 and say why it is called KEY CHECK: that is the minute somebody finds out whether the dish is on time.

### Slide 6: Sticky notes into blocks

- One step per sticky note.
- For each note: how long does this take, and **who does it.**
- Place every note into a block. Move them until no block is empty for anybody. Notes: 12 minutes. Ask one question per station: “What is happening in your 15 to 20 block? Show me.”

### Slide 7: The critical path

- **Point to the step that, if it is late, makes everything after it late.**
- That is your critical path.
- The pasta station’s is the water boiling. Nothing else can happen until it boils, so the water goes on in block one, before anyone washes a cup.
- Circle it on your poster in a second color.

### Slide 8: The slip test

- If your critical path step starts **five minutes late:**
- It finishes at minute \_\_\_ instead of \_\_\_.
- The dish is ready at minute \_\_\_.
- **\*\*We eat at minute \_\_\_.\*\***
- Circle: we still eat inside the period / we do not. Notes: A real time, not “we would be late.”

### Slide 9: Three shortcuts, written before you need them

1. **Split the batch.** Two smaller pots or trays cook faster than one big one.
  2. **Move it.** Second oven, toaster oven, microwave for part of it.
  3. **Drop the last optional step.** No second cheese layer, no garnish.
- “Work faster” is not a shortcut.
  - **The rule: the clock is not negotiable, the presentation is.**

### Slide 10: The station card, four parts

1. Ingredients for this station
2. **The allergen box** (the card is not approved without it)
3. **Ten numbered steps, one action per step**

4. Safety reminders, and “what does done look like” Notes: One action per step. “Boil the water, add the pasta, and set the timer” is three steps, not one.

### Slide 11: What “done looks like” actually means

- Not: “it tastes good.”
- Yes: “pasta is soft, not crunchy.” “Sauce is mixed evenly.” “Cheese is bubbling and the edges are brown.”
- Yes: “Burner OFF. Station wiped. Pot rinsed.” Notes: Four checks, each one something you can see.

### Slide 12: The paired read

- Trade cards with the station next door.
- They write **one question the card could not answer.**
- Fix that step before the bell.
- **The test: could a substitute teacher run this station from your card alone?**

### Slide 13: Post to the class master grid

- One person per station copies the eight blocks into your column.
- Then we read the grid across, block by block, out loud.
- We stop at the collisions: two stations on the oven, two stations needing me at the same minute, everybody plating at once.
- We fix them in red. Notes: Closure: each station reads “Our critical path is , **it starts at minute** , and if it slips we eat at \_\_\_\_.” Then the whole class reads the 30 to 35 block across all four columns. Posters stay up through Lesson 6.17.